

Cross-Technology Signal Detection and Jamming Attack for Heterogeneous Internet of Things

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Abstract—With the rapid development of the Internet of Things (IoT) and 5 G networks, the coexistence of diverse wireless protocols (e.g., Wi-Fi, ZigBee, and BLE) in the 2.4 GHz band has created new opportunities and security challenges for Cross-Technology Communication (CTC). This paper proposes an intelligent jamming attack framework based on channel state information (CSI) to identify and disrupt CTC communications, where an adversary can leverage fine-grained CSI to build a closed-loop attack. We design a dual-branch LSTM neural network model integrated with an attention mechanism, achieving high-accuracy classification of Wi-Fi, ZigBee/BLE, and CTC signals with an accuracy exceeding 96%. Furthermore, we model the jamming decision process as a Markov Decision Process (MDP), employing LSTM to predict signal behavior and leveraging a Deep Q-Network (DQN) for slot-level intelligent jamming optimization. The proposed system is implemented on an AC-1200 Wi-Fi network interface card and a TI CC2650 platform, with its real-time performance and jamming effectiveness validated in a real-world 2.4 GHz testbed. Experimental results demonstrate that our approach significantly reduces the throughput of ZigBee/BLE communications while maintaining high attack efficiency and stealth, offering new insights into the security of heterogeneous wireless networks.

Index Terms—Cross-technology communication, channel state information, jamming attack, heterogeneous wireless networks.

I. INTRODUCTION

WITH the explosive growth of IoT devices and the gradual commercialization of 5 G networks, wireless communication environments are rapidly evolving from homogeneous to highly heterogeneous systems [1], [2]. In this context, a wide range of wireless protocols, including high-throughput Wi-Fi, low-power ZigBee, and BLE, have been extensively deployed within the unlicensed 2.4 GHz Industrial, Scientific, and Medical (ISM) band [3], [4]. This co-existence has become foundational for numerous mission-critical applications such as smart

homes [5], industrial automation [6], wearable health monitoring [7], and smart cities [8]. However, significant differences in modulation schemes, channel widths, and medium access control mechanisms among these protocols pose considerable challenges in terms of interoperability, spectrum contention, and security assurance [9].

Traditional approaches to achieving interoperability typically rely on protocol gateways or bridge devices that perform translation and coordination at higher communication layers [10]. While functional, these solutions introduce nonnegligible communication latency and processing overhead, require dedicated hardware, and often create single points of failure, limits that are especially problematic in distributed and resource-constrained IoT deployments.

To address the need for direct communication between heterogeneous devices, Cross-Technology Communication (CTC) [11], [12] has emerged as a promising alternative. CTC bypasses the constraints of unified protocol stacks by embedding information directly into the physical-layer features of existing transmissions. Although these perturbations are virtually imperceptible to native receivers, they can be decoded by non-conforming devices equipped with compatible demodulation logic, thereby enabling inter-protocol communication without modifying hardware or breaking protocol compliance. As a result, CTC significantly reduces system complexity and energy consumption while enhancing cross-protocol flexibility.

However, many conventional security mechanisms designed for homogeneous IoT environments fail to scale effectively in heterogeneous deployments, thereby limiting their applicability to cross-technology scenarios [13]. The protocol transparency and low-layer operation of CTC introduce substantial security vulnerabilities. Most existing CTC implementations lack standardized encryption, authentication, and integrity verification mechanisms. Consequently, adversaries can exploit these deficiencies to eavesdrop, inject forged messages, or launch denial-of-service (DoS) attacks, often while remaining undetected [14]. In practice, such attacks may enable the injection of false control signals, disruption of coordination among IoT nodes, or even hijacking of communication flows, posing serious threats to the operational reliability of heterogeneous networks.

In this context, Channel State Information (CSI) has emerged as a critical enabler for both analyzing and manipulating CTC. CSI provides fine-grained representations of the wireless channel by capturing the amplitude and phase response of each OFDM subcarrier as it propagates through multipath environments [15]. Compared to coarse metrics such as Received Signal

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Strength Indicator (RSSI) or energy detection, CSI offers significantly higher resolution and sensitivity, making it well suited for identifying protocol, specific physical-layer fingerprints embedded in CTC transmissions. CTC often encodes information by subtly modulating a small subset of subcarriers. While these perturbations are nearly imperceptible in RSSI measurements, they leave distinguishable patterns in the CSI time–frequency domain. For example, Wi-Fi transmitters that emulate ZigBee, compatible preamble structures may induce periodic phase shifts or amplitude anomalies in specific subcarriers [16]. Similarly, BLE's frequency, hopping behavior may appear as predictable fluctuations in CSI over time [17]. By modeling CSI as a time series and analyzing its multidimensional variations, adversaries can not only detect ongoing CTC activity, but also infer transmission timing, duration, and spectral characteristics, enabling precise, adaptive, and covert attacks. This capability elevates CSI to a powerful asset for intelligent adversaries. By continuously monitoring CSI and applying sequence modeling techniques, an attacker can predict the timing of CTC transmissions, identify active subcarriers, and determine the emulated protocol. These insights enable selective jamming, signal spoofing, and the injection of crafted CTC frames, each precisely tailored to the structure and behavior of the target communication.

Despite growing interest in CTC mechanisms and their security implications, existing work rarely investigates how adversaries can construct adaptive, data-driven attack strategies based on fine-grained CSI features. Realizing such attacks requires overcoming several open challenges, such as real-time processing of high dimensional CSI, maintaining generalization across diverse channel conditions, and optimizing interference strategies under energy and latency constraints.

In this paper, we fill this gap by presenting a comprehensive framework for intelligent jamming attacks on CTC enabled links. Specifically, our contributions are threefold:

- We design a dual-branch recurrent neural network that processes amplitude and phase CSI in parallel with an attention mechanism, achieving over 96% classification accuracy for Wi-Fi, ZigBee/BLE, and CTC across varying channel conditions.
- We formulate the jamming process as an MDP over 1 ms slots, using LSTM to predict victim activity and DQN to learn efficient jamming decisions balancing effectiveness and energy cost.
- We implement the full system on AC1200 Wi-Fi NICs and TI CC2650 platforms. Experiments on a 2.4 GHz testbed validate sub-millisecond CSI processing and real-time operation within 1 ms slots.

The remainder of this paper is organized as follows. Section II reviews related work on jamming attacks and defenses in IoT and CTC networks. Section III presents the motivation, highlighting both the challenges and opportunities in securing heterogeneous wireless networks. Section IV describes the system overview and attack scenario. Section V details the CSI-based monitoring and detection model using a dual-branch LSTM with an attention mechanism. Section VI presents the LSTM-DQN-based jamming attack model. Section VII describes the experimental setup and evaluation results. Section VIII discusses the

implications and limitations of the proposed framework. Section IX concludes the paper.

II. RELATED WORK

A. Jamming Attacks and Detection in IoT Networks

Jamming attacks are a persistent threat to IoT networks due to their broadcast nature and resource constraints. Comprehensive surveys by [18] and [19] classify jamming attacks across various wireless networks, including IoT related systems, and highlight the resource constraints of IoT devices as a limiting factor for antijamming techniques. Reactive jamming, in particular, is difficult to detect because it activates only during legitimate transmissions. To address this, a detection scheme based on threshold and RSSI analysis, CDCA, has been proposed, which significantly improves network throughput and energy efficiency in attack scenarios [20].

Intelligent and adaptive jamming strategies are also emerging. A Markov chain based model demonstrates how attackers can optimize energy usage while maintaining high jamming success rates, outperforming basic jamming approaches [21]. Coalition based attacks further exploit compromised IoT nodes that coordinate based on prior network knowledge to amplify disruption [22]. On the defense side, machine learning methods are increasingly adopted. The JCED framework classifies jammer types and applies targeted countermeasures, achieving better throughput and energy savings than traditional detection algorithms [23]. Federated learning has been used to detect multiple types of jamming while preserving user privacy, reaching accuracy up to 99% [24]. New vulnerabilities are also exposed in federated learning itself, where attackers can jam uplink or downlink transmissions, significantly affecting model convergence [25]. Physical layer solutions such as the Number of Jammed Slots (NJS) metric improve detection and localization speed and accuracy in multi-hop IoT networks [26]. To support the evaluation, ns-3-based jamming modules allow the simulation and comparison of various strategies under realistic conditions [27]. Broader overviews highlight risks in autonomous aerial vehicle (AAV) [28] and 5 G networks [19], emphasizing the need for robust and adaptive defense mechanisms.

B. Cross-Technology Communications

Existing CTC techniques can be broadly classified into two categories: packet-level CTC and physical-layer CTC.

Packet-level CTC encodes control information into coarse-grained transmission features, such as packet size (e.g., Esense [35], HoWiES [36]), packet scheduling (e.g., FreeBee [37], DCTC [38], Gap Sense [39], C-Morse [40]), or transmission energy (e.g., WiZig [41], StripComm [42]). Furthermore, some approaches take advantage of Wi-Fi's ability to directly acquire CSI by embedding data into CSI, as seen in works such as ZigFi [43], AdaComm [44], cChirp [45], and DopplerFi [46]. These methods do not modify the underlying physical waveform, making them non-intrusive and compatible with commodity devices. Packet-level CTC offers simplicity in deployment and protocol transparency. However, due to their reliance on

TABLE I
COMPARISON WITH REPRESENTATIVE PRIOR STUDIES IN CTC SECURITY AND INTELLIGENT JAMMING

Work	Scenario	Core Technique	Detection	Jamming	Main Difference from This Work
Hide and Seek [29]	Wi-Fi forging ZigBee packets	Packet forgery and higher-order	No	No	Packet injection/control in ZigBee, rather than CTC links.
Wi-Attack [30]	Wi-Fi spoofing BLE broadcasts	Cross-technology signal simulation	No	No	Targets BLE location spoofing, without fine-grained CSI modeling.
AuthCTC [31]	Authentication vs. CTC signals	Physical-layer similarity measurement	Partial	No	Improves signal authentication, but does not model transmission timing.
DeepJam [14]	ZigBee communication under CTI	DQN-based jamming using CTI sequences	No	Yes	Learns jamming timing, but does not first identify CTC signal.
JamCloak [32]	Multiple CTC protocols	DNN-based CTC protocol classification	No/limited	Partial	Supports CTC-oriented interference, but lacks a unified CSI-based identification.
LSTM-based jammer [33]	jamming under dynamic channels	LSTM-based CSI prediction with jamming	Partial	Yes	Uses CSI prediction for jamming decisions, but does not address CTC signal.
LSTM-based detection [34]	ZigBee emulation detection	LSTM-based learning of chipset error patterns	Partial	No	Detects unauthorized, but does not further integrate prediction.
Our work	Heterogeneous CTC links	Dual-branch LSTM-attention for jamming	Yes	Yes	Provides a unified CSI-driven closed-loop attack framework.

sparse metadata and indirect modulation, these approaches typically achieve limited throughput, often constrained to a few kilobits per second (kbps).

In contrast, a Physical-layer CTC synthesizes waveform-level signals that align with the demodulation logic of the target technology, enabling significantly higher data rates. The seminal WEBee [16] project demonstrated that Wi-Fi devices can emulate ZigBee signals by embedding Direct Sequence Spread Spectrum (DSSS) modulated payloads within Orthogonal Frequency Division Multiplexing (OFDM) frames. Subsequent research extended waveform emulation to other wireless technologies, including Wi-Fi to BLE (e.g., BlueFi [47], WiBLE [48], Unify [49], WiBeacon [50]), Wi-Fi to LoRa (e.g., WiRa [51], Wi-Lo [52]), and BLE to ZigBee (e.g., BlueBee [53]). More recently, innovative approaches have integrated neural networks into CTC. For instance, Xituxi [54] treats payloads from different signals as distinct language texts, leveraging large language models to map payloads across heterogeneous technologies. Similarly, NNCTC [55] employs deep learning to train neural networks that autonomously learn end-to-end physical-layer waveform mappings between diverse standards.

C. Jamming Attacks in CTC

Traditional jamming techniques in CTC systems typically rely on energy flooding or random noise injection. Although these methods are straightforward to implement, they generally suffer from low energy efficiency, large interference footprints, and limited stealth. To overcome these limitations, recent studies have begun to investigate the vulnerabilities inherent in CTC protocols. For example, Hide and Seek [29] demonstrates that Wi-Fi devices can forge ZigBee packets to covertly control ZigBee nodes and introduces a detection method based on higher-order constellation statistics. Wi-Attack [30] presents a method for location spoofing by simulating BLE iBeacon broadcasts using Wi-Fi devices. AuthCTC [31] further enhances detection and authentication by measuring the similarity between emulated and genuine signals at the physical layer.

In addition to protocol-level vulnerabilities, neural network based intelligent jamming approaches have gained increasing

attention. DeepJam [14] employs a DQN to learn temporal features from Cross-Technology Interference (CTI) sequences and selectively jam ZigBee transmissions during predicted time slots. JamCloak [32] proposes a classification scheme for existing CTC protocols and uses deep neural networks to extract signal features, enabling accurate detection and effective interference against various mainstream CTC schemes. Similarly, [33] explores multiple jamming strategies, including continuous, passive, and learning assisted attacks. It applies a LSTM network to predict CSI, guiding optimal jamming decisions. Furthermore, [34] utilizes inherent error patterns in ZigBee chipsets and applies LSTM based learning to capture emulation anomalies, significantly improving unauthorized signal detection and authentication. The comparison with representative prior studies in CTC security and intelligent jamming is shown in Table I.

III. MOTIVATION

A. Background

The rapid proliferation of heterogeneous IoT devices has fostered innovative CTC paradigms. For instance, recent works have leveraged CTC to enable pervasive internet connectivity for commodity IoT devices without hardware gateways [56], achieve high-precision time synchronization for wearables [57], and expand BLE location based services using ubiquitous Wi-Fi infrastructure [50]. These applications demonstrate that CTC has evolved from a theoretical concept into a fundamental service provider for synchronization, positioning, and network management.

Fig. 1 shows the increasing reliance on the physical layer for cross-protocol interaction introduces severe security vulnerabilities. Since CTC emulation creates unique signal patterns in the CSI, a passive adversary can exploit these fine-grained fingerprints to identify sensitive control flows or synchronization pulses. By recognizing these patterns, an intelligent attacker can launch highly selective jamming, disrupting critical services like time synchronization or emergency commands with minimal energy expenditure.

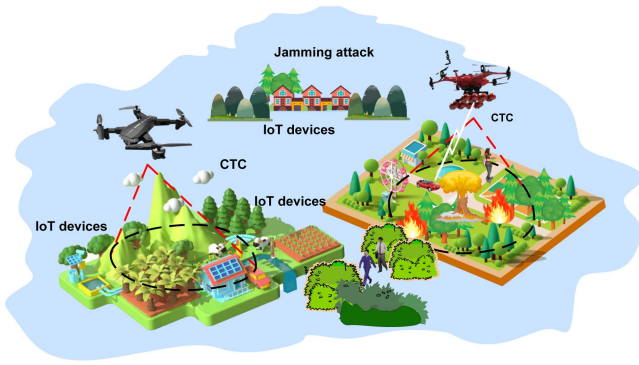


Fig. 1. The CSI based attacks for a heterogeneous IoT scenario, where a large number of heterogeneous IoT devices are used, while traditional attacks only occur on homogeneous devices. The attack described in this paper can target heterogeneous devices.

B. Challenge

As wireless networks evolve toward greater heterogeneity and spatial density, physical-layer security threats arising from cross-technology coexistence have emerged as critical barriers to the reliable operation of the IoT. Traditional defense mechanisms predominantly rely on higher-layer protocols (e.g., MAC or network layers) for threat detection and mitigation. However, these approaches are inherently limited in their ability to bridge the gap between heterogeneous communication technologies or to capture physical-layer artifacts of cross-technology interference. This presents two core challenges: insufficient signal perception granularity and heightened stealth of adversarial actions.

For instance, an attacker may struggle to accurately differentiate between Wi-Fi data frames and ZigBee control packets [58]. Naively applying wideband interference not only risks simultaneously triggering backoff mechanisms across multiple protocols but also diminishes attack efficiency when target devices promptly adapt via dynamic channel hopping. Therefore, a fundamental problem arises: how to achieve fine-grained recognition of cross-technology signal features to enable targeted, adaptive jamming attacks that enhance both stealth and disruptive capability. Current research into cross-technology security threats faces two critical limitations:

1) *Heterogeneous Signal Feature Extraction*: Conventional identification methods, such as those based on RSSI or spectral energy aggregation, lack the resolution to distinguish between disparate modulation schemes and protocol stacks. For example, although Wi-Fi (OFDM based) and ZigBee (DSSS based) transmissions may exhibit similar energy distributions in the time–frequency domain, their CSI reflects distinct multipath characteristics. Coarse grained features such as RSSI are prone to misclassification under signal superposition or multipath fading, rendering them ineffective for precise classification. Moreover, these methods are poorly suited to environments with dynamic channel variations, thereby limiting their applicability in complex heterogeneous networks.

2) *Stealth and Adaptivity of Jamming Attacks*: Due to the time-varying nature of wireless channels, effective jamming

requires continuous monitoring of channel conditions and real-time adjustment of interference parameters. This imposes stringent demands on the attack system, including low-latency decision-making and high-throughput data processing. Furthermore, to evade traditional detection mechanisms, adversaries must embed attack signals within ambient noise or legitimate traffic patterns, complicating the design of both attack and defense strategies.

These challenges stem from the inherent interaction between cross-technology interference and the dynamics of the channel at the physical layer [14]. Traditional detection frameworks absent fine grained analysis of CSI and related features struggle to capture subtle footprints of malicious activity. Moreover, existing jamming models based on random or broadband noise lack the adaptability required for precise, environment-sensitive attacks.

C. Opportunities

Despite the aforementioned challenges, the use of CSI presents a promising paradigm shift for both cross-technology attack strategies and defense mechanisms. As a fine-grained physical-layer representation of the wireless channel, CSI inherently transcends protocol heterogeneity and captures the multidimensional (time, frequency, and spatial) characteristics of signal propagation. This offers three principal opportunities:

1) *Fine Grained Activity Detection Via CSI*: CSI enables precise differentiation between communication standards by capturing protocol-specific frequency domain characteristics [59]. For example, ZigBee’s DSSS signals typically exhibit a lower ranking structure in the CSI subcarrier correlation matrix compared to Wi-Fi’s OFDM signals. Similarly, BLE communications produce unique periodic patterns in the time-domain auto-correlation of CSI because of their frequency hopping behavior. These distinctive traits can be exploited to construct robust classification models capable of identifying specific transmission types such as Wi-Fi data frames or ZigBee control messages—even in dense co-existence scenarios. Such fine-grained detection not only enhances situational awareness, but also enables adversaries to isolate and target specific CTC events for selective interference.

2) *Low-Power, Precision Jamming*: The channel awareness enabled by CSI facilitates energy efficient and targeted jamming strategies. By analyzing CSI, an adversary can identify the exact subcarriers and spectral bands utilized by a target device, focusing interference solely within those regions. This selective jamming significantly reduces energy expenditure while improving attack precision. Moreover, CSI can be used to predict short term channel evolution, such as occupancy bursts or multipath fluctuations, allowing the attacker to time their actions during critical transmission intervals of the victim. This not only amplifies the efficacy of the attack, but also minimizes the probability of detection.

3) *Improved Stealth Through Environmental Mimicry*: CSI inherently reflects multipath propagation effects, granting attackers a detailed understanding of the physical environment. Leveraging this information, adversaries can synthesize

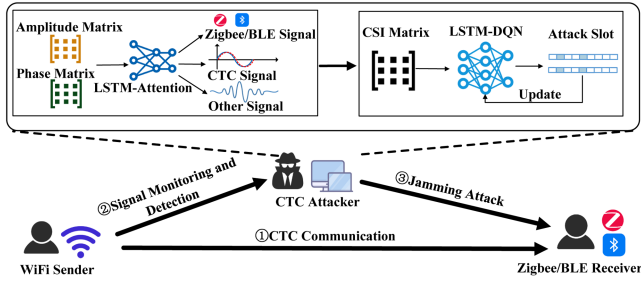


Fig. 2. System overview.

interference signals that emulate the statistical properties of ambient noise or environmental fluctuations. By blending with the natural variability of the wireless channel, such interference can evade conventional detection techniques based on abrupt RSSI changes or abnormal spectral energy patterns. Consequently, CSI-guided attacks raise the stealth threshold, challenging existing detection schemes, and necessitating the development of more advanced defense strategies.

CSI serves as a powerful enabler for both offensive and defensive mechanisms in heterogeneous wireless environments. By facilitating high resolution signal identification and supporting dynamic, adaptive jamming strategies, CSI not only exposes previously unobservable threats but also provides the foundation for building next generation security solutions. As CTC continue to proliferate in IoT ecosystems, harnessing the full potential of CSI is imperative to advance the state of physical layer security.

IV. OVERVIEW

Building on insights into CSI driven vulnerabilities and motivation for adaptive jamming, this section presents the overall design of our proposed attack framework, detailing its system architecture and operational settings in heterogeneous CTC environments. This system proposes an intelligent jamming attack framework that leverages CSI time series modeling to disrupt CTC in heterogeneous wireless networks operating in the 2.4 GHz band. The overall architecture, as depicted in Fig. 2, integrates two primary modules: a monitoring and detection model and a jamming attack model. These components work in tandem to detect high precision CTC traffic and execute energy efficient adaptive interference strategies.

The monitoring and detection model employs a dual-branch LSTM network augmented with an attention mechanism. This model processes CSI data by separately analyzing amplitude and phase streams, extracting fine grained features to classify incoming signals as Wi-Fi, ZigBee/BLE, or CTC. The attention mechanism improves classification by emphasizing distinctive sub-carrier patterns, achieving an accuracy exceeding 96% across diverse channel conditions. This robust detection capability forms the foundation for identifying CTC transmissions in real time.

The jamming attack model formalizes the interference strategy as an MDP over discrete 1-millisecond time slots. Using a lightweight LSTM to predict the probability of transmission

of the victim based on historical CSI data, enabling proactive anticipation of CTC activity. A DQN then optimizes the jamming decision, whether to transmit noise or remain idle, balancing throughput reduction with energy efficiency and stealth. This adaptive approach ensures precise interference targeting key transmission slots, minimizing redundant actions. By combining advanced deep learning with reinforcement learning, this framework offers a novel, unified approach to CTC jamming, outperforming traditional rule-based strategies in adaptability and effectiveness.

V. CHANNEL MONITORING AND DETECTION MODEL

The proliferation of heterogeneous wireless technologies in the 2.4 GHz band, coupled with the vulnerabilities of CTC, requires robust mechanisms to detect and classify diverse signal types accurately. The primary challenge lies in distinguishing subtle CTC signal fingerprints from native protocol transmissions under dynamic channel conditions. This chapter introduces a monitoring and detection model that leverages CSI and a dual-branch recurrent neural network with an attention mechanism to achieve high-precision signal classification, addressing the critical need for real-time situational awareness to secure CTC enabled networks.

A. Preliminary

This section first delineates the entity system involved in our study and then introduces the attack scenario. Consider a setup in which a legitimate Wi-Fi sender engages in CTC with a ZigBee or BLE device, as depicted in Fig. 2.

1) *Wi-Fi Sender*: The Wi-Fi sender is an authentic IEEE 802.11 device that initiates CTC with the ZigBee/BLE receiver. By reverse engineering the Wi-Fi PHY layer, this sender transmits bespoke CTC waveforms over the standard Wi-Fi protocol. Upon reception, the ZigBee/BLE device interprets these signals as legitimate, despite their origin in the Wi-Fi domain.

2) *ZigBee/BLE Receiver*: The ZigBee/BLE receiver serves as the target device. In the CTC framework, its native security features are selectively disabled to allow reception of cross-technology signals. Consequently, the receiver processes both communications from the authorized Wi-Fi sender and those injected by a malicious adversary as valid data frames.

3) *CTC Attacker*: The CTC attacker is a malicious device capable of Wi-Fi that can be manipulated to exploit the CTC mechanism. It continuously monitors the Wi-Fi spectrum for signatures indicative of CTC activity. Once the attacker detects an ongoing exchange between the Wi-Fi sender and the ZigBee/BLE receiver, it analyzes the extracted signal characteristics (e.g., preamble structure, modulation patterns) and synthesizes identical waveforms. By transmitting these spoofed CTC signals in predicted time intervals with matching frequency and symbol timing, the attacker aims to disrupt or hijack legitimate CTC.

Attack Scenario: In the described scenario, the Wi-Fi sender transmits CTC signals to the ZigBee/BLE receiver. Currently, the CTC attacker listens passively for Wi-Fi frames containing CTC information. Upon detection, it replicates and injects counterfeit CTC signals into the channel, thereby corrupting the

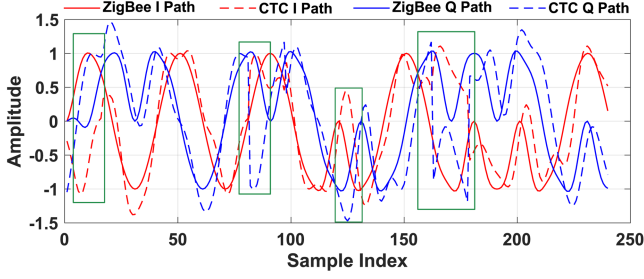


Fig. 3. Time-domain comparison of ZigBee and emulated CTC signals on I and Q components. Highlighted regions (green boxes) indicate areas where amplitude deviations occur between the original and emulated signals.

original communication or delivering malicious payloads that the ZigBee/BLE receiver erroneously processes as authentic.

B. CSI Analysis

CSI provides a fine grained representation of wireless channel characteristics, far beyond what is offered by conventional metrics such as RSSI. CSI captures how signals propagate from the transmitter to the receiver over each subcarrier in an OFDM system, reflecting multipath fading, attenuation, interference, and phase distortion. The CSI for a Wi-Fi signal over N subcarriers can be modeled as a complex gain:

$$H(f_k) = |H(f_k)|e^{j\angle H(f_k)}, \quad k = 1, 2, \dots, N, \quad (1)$$

where $\|H(f_k)\|$ and $\angle H(f_k)$ denote the amplitude and phase of the channel frequency response of the k -th subcarrier, respectively. Given that Wi-Fi operates over a much wider bandwidth (20 MHz) compared to ZigBee and BLE (both 2 MHz), its CSI inherently captures spectral information that overlaps with the narrower-band transmissions. For instance, when Wi-Fi operates on channel 6 (centered at 2437 MHz), subcarriers indexed by $k = -6, -5, -4$ and $k = 25, 26, 27$ align with ZigBee channels 17 and 19. Focusing on these overlapping subcarriers allows the system to isolate and analyze the impact of cross-technology signals on the Wi-Fi CSI.

In physical-layer CTC, attackers emulate ZigBee or BLE signals by subtly modulating selected Wi-Fi subcarriers, introducing fine grained perturbations in phase and amplitude. Although these emulated signals closely mimic the target modulation, perfect reproduction is inherently limited by hardware imperfections, timing mismatch, and signal shaping inaccuracies. As a result, such imperfections, though imperceptible at the RSSI or waveform level, leave distinct and protocol-specific anomalies in the CSI domain. These manifest as abnormal amplitude spikes or phase oscillations on subcarriers overlapping with ZigBee or BLE bands, such as channels 17 and 19 when Wi-Fi uses channel 6.

Fig. 3 illustrates this phenomenon by comparing the I/Q baseband traces of native ZigBee signals (solid lines) and their CTC emulated counterparts (dashed lines). Fig. 4 compares the phase shifts of native ZigBee signals (solid lines) and their CTC emulated counterparts (dashed lines). Despite their overall similarity, the highlighted regions reveal discrepancies in amplitude

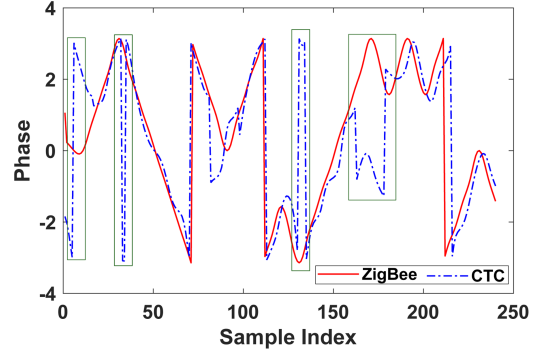


Fig. 4. Phase comparison of ZigBee and emulated CTC signals. Highlighted regions (green boxes) indicate areas where phase deviations occur between the original and emulated signals.

and phase, which translate into observable distortions in the CSI matrix. These emulation induced variations are consistent and repeatable across transmissions, forming recognizable patterns that can be exploited by our LSTM-Attention model. By focusing CSI feature extraction on overlapping subcarriers, the model learns to distinguish real from emulated signals with high fidelity, even under fluctuating propagation conditions. This demonstrates that CSI not only offers superior resolution for detecting cross-technology activities but also acts as a robust physical-layer fingerprint for uncovering subtle signal emulation artifacts.

C. LSTM-Attention for Signal Classification

The initial phase of our attack framework is dedicated to continuous monitoring and classification of CTC traffic in real time. To achieve this, the adversary persistently samples CSI from the wireless environment and applies signal preprocessing to extract fine-grained, discriminative features that reflect the presence of CTC.

We propose a hybrid deep learning architecture that integrates dual-branch LSTM networks with an attention mechanism to enable robust and accurate CTC detection. Each LSTM branch—one dedicated to CSI amplitude and the other to CSI phase—comprises two stacked layers that are capable of capturing long-range temporal dependencies, thereby preserving sequential dynamics in the input CSI streams. At each time step, the hidden state outputs are projected through fully connected layers to produce subcarrier-aligned feature vectors, effectively encoding signal characteristics at the subcarrier level.

Subsequently, an attention module computes a relevance score for every time-step/subcarrier pair. These scores are normalized into attention weights and used to perform a weighted summation over the feature vectors. This mechanism adaptively emphasizes spectral-temporal patterns that are indicative of BLE/ZigBee induced perturbations, while suppressing ambient channel noise and noninformative fluctuations. To illustrate this architecture, we provide an example based on ZigBee signals.

Fig. 5 illustrates the architecture of the proposed LSTM-Attention network. The model consists of two parallel branches that separately process the amplitude and phase

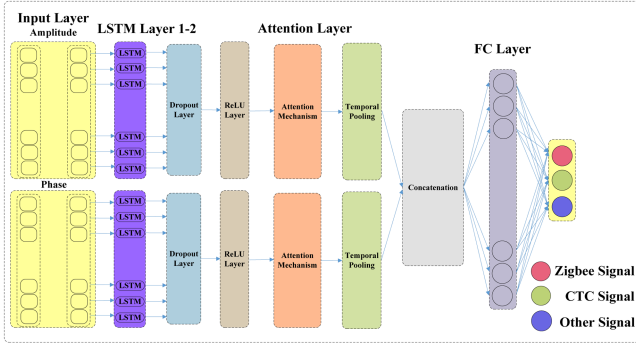


Fig. 5. The architecture of LSTM-Attention network.

components of the input CSI stream. Each branch includes two LSTM layers for temporal sequence modeling, followed by a fully connected layer that maps the LSTM output to a per-subcarrier representation. An attention module in each branch computes relevance scores across time and frequency domains, enabling the model to focus on subcarriers that carry meaningful modulation signatures introduced by CTC events. The attention weighted feature vectors from both branches are then concatenated into a unified representation. To further improve generalization under different antenna and propagation configurations, the model integrates multi-antenna spatial diversity through a global average pooling layer applied across the temporal dimension of all receive chains. This fusion mechanism aggregates the dynamics of the cross-channel signal while retaining interference, specific fingerprints. The resulting feature vector is subsequently passed to the classification head for final decision making.

To exploit complementary information from both CSI modalities, the outputs from the amplitude and phase branches are fused as a single feature vector. To enhance robustness across dynamic environments, particularly in the presence of multipath fading and device diversity, we incorporate spatial information via a fusion layer that performs global average pooling over the temporal dimension across all available antennas.

Finally, the fused feature representation is input to a classification head consisting of a fully connected layer followed by a Softmax activation function, which produces posterior probabilities for the three target classes: native Wi-Fi, ZigBee, and CTC signals. Formally, the probability p_i that the input sample belongs to class i is computed from the corresponding output logit z_i , such that: $p_i = e^{z_i} / \sum_{j=1}^3 e^{z_j}$. To train the network, we minimize the standard cross-entropy loss over all training samples:

$$\mathcal{L}_{CE} = - \sum_{n=1}^N \sum_{i=1}^3 y_{(n,i)} \log \hat{y}_{(n,i)}, \quad (2)$$

where N is the total number of training instances, $y_{(n,i)}$ is the ground-truth one-hot label for sample n , and $\hat{y}_{(n,i)}$ is the predicted probability for class i . To prevent overfitting and promote generalization across channel conditions, we apply dropout regularization with a rate of 0.3 within the classification

layers. All network hyperparameters, including LSTM hidden dimensions, attention dimensions, learning rate, and batch size, are selected through grid search and five-fold cross-validation.

The accuracy of the trained network is determined by its classification performance on a held-out test set that is not used during parameter updating. Specifically, after constructing the labeled CSI dataset, we randomly divide the samples into 70% for training and 30% for testing. The training subset is used to optimize the network parameters by minimizing the cross-entropy loss, while the testing subset is reserved exclusively for evaluating the final model after training is completed. In this work, classification accuracy, defined as the ratio of correctly predicted samples to the total number of test samples, is adopted as the primary evaluation metric because the task is formulated as a multi-class classification problem. During inference, the network outputs the posterior probability of each class through the Softmax layer, and the predicted label is taken as the class with the highest probability. The final reported accuracy is then obtained by comparing these predicted labels with the corresponding ground truth labels on the unseen test data.

To ensure that the reported accuracy is reliable rather than the result of a particular parameter setting, the key hyperparameters of the network, such as the hidden dimension, attention dimension, learning rate, and batch size, are selected through grid search combined with five-fold cross-validation on the training set. This procedure helps identify a robust configuration that generalizes well across different subsets of the data. In addition, dropout regularization is introduced during training to mitigate overfitting and improve the generalization capability of the network. Therefore, the reported accuracy is not determined by training performance alone, but by the model's ability to correctly classify previously unseen CSI samples under the selected hyperparameter setting. This evaluation strategy provides a more objective measure of the effectiveness and practical applicability of the trained network.

Empirical results show that our model achieves a classification accuracy of over 96% on real-world CSI datasets, outperforming conventional machine learning baselines. This high accuracy enables precise detection of CTC transmissions even under challenging multi-protocol coexistence scenarios, thereby providing a reliable foundation for subsequent jamming strategies.

We selected the LSTM architecture because the CSI observations used in our system are inherently sequential and exhibit strong temporal correlation across consecutive transmission slots. In our scenario, the received CSI is not an independent snapshot at each instant; instead, the amplitude and phase values evolve continuously with channel dynamics, transmitter behavior, and cross-technology interference patterns. As a result, both CTC signal detection and jamming-slot prediction depend not only on the current CSI measurement, but also on the temporal evolution of CSI over multiple previous slots. This characteristic makes the problem fundamentally different from a static classification task and motivates the use of a sequence-learning model.

Compared with conventional feed-forward networks or static classifiers that process each CSI sample independently, LSTM is able to preserve historical information through its memory

cell and gating mechanism, allowing it to capture both short-term fluctuations and longer-term temporal dependencies in the wireless channel. This capability is particularly important in our task because CTC related perturbations are often weak, transient, and embedded in normal channel variations. By modeling the correlation among consecutive CSI samples, LSTM can better distinguish structured CTC induced patterns from random noise or ordinary channel fading. In addition, compared with standard RNNs, LSTM is more effective in mitigating gradient vanishing and gradient explosion problems, which improves training stability when learning from longer CSI sequences.

Another reason for selecting LSTM is that the two main objectives of our framework, namely CTC signal classification and proactive jamming decision, are both closely related to temporal prediction. For signal classification, the detector needs to identify whether the observed CSI sequence corresponds to normal ZigBee/BLE activity, Wi-Fi-based CTC behavior, or non-CTC background conditions. For jamming optimization, the attacker must infer the transmission tendency of the victim node in the upcoming slot based on previously observed CSI sequences. In both cases, the network must learn the temporal context and inter-slot dependency rather than rely on isolated per-slot features. Therefore, LSTM provides a unified and natural architecture for these two modules.

We also considered the practical deployment requirement of real-time operation. The decision process in our system is performed on short time scales, such as 1 ms slots, which requires the model to achieve a balance between representation capability and computational complexity. LSTM offers this tradeoff well: it is sufficiently expressive to model time-varying wireless behaviors, while still remaining lightweight enough for efficient inference compared with more complex sequence models. Therefore, the choice of LSTM is driven by the sequential nature of CSI, the need to capture temporal dependencies for both detection and jamming prediction, and the practical requirement of stable and real-time operation in dynamic wireless environments.

VI. JAMMING ATTACK MODEL

A. Attack Model

To rigorously analyze the proposed jamming framework, we define a concrete attack model specifying the adversary's goals, capabilities, knowledge, and constraints in heterogeneous IoT environments. We formulate CTC jamming as a fine grained sequential decision problem that is compatible with real wireless constraints. In particular, the attacker must act under millisecond-level timing, partial observability from CSI, and a multi-objective tradeoff among disruption effectiveness, energy expenditure, and stealth. This formulation distinguishes our problem from conventional reactive jamming or protocol-specific interference models, and motivates the use of a learning based controller.

Attacker Goals: The primary objective is to disrupt CTC between a Wi-Fi transmitter and a ZigBee or BLE receiver. The attacker aims to accurately identify CTC activity and strategically transmit interference signals to reduce throughput and increase packet error rates at the victim receiver.

Attacker Capabilities:

- *Real-time CSI Monitoring:* The attacker can continuously capture CSI from ambient Wi-Fi transmissions using commercial Wi-Fi Network Interface Cards (NICs) that expose CSI data.
- *Protocol Knowledge:* The attacker is assumed to have knowledge of Wi-Fi, ZigBee, and BLE PHY-layer characteristics, including their timing, spectrum usage, and modulation behavior.
- *Jamming Transmission:* The attacker can transmit customized waveforms using commodity Wi-Fi hardware with timing and frequency control to overlap with CTC transmissions.
- *Computation Resources:* The attacker is equipped with sufficient computational capability to run lightweight LSTM and DQN models in real-time for slot-based prediction and decision-making.

Attacker Knowledge:

- The attacker does not require access to encryption keys, traffic contents, or protocol headers. The attack operates purely at the physical layer using observable CSI features.
- The attacker is aware of the CTC encoding structure and the timing behavior of the legitimate sender and receiver.

Attacker Constraints:

- *Non-intrusiveness:* The attacker is external and does not compromise any device. It passively observes the channel and injects interference without modifying network protocols.
- *Power and Stealth Limits:* The jamming signal must conform to regulatory power constraints and minimize detectability by avoiding constant or wideband interference.
- *Environmental Variability:* The attack must remain effective under time-varying channel conditions, mobility, and multipath fading, which affect CSI measurements.

This defined model reflects a realistic and capable physical-layer adversary in smart home, industrial, and urban IoT deployments. It supports reproducibility and highlights the generalizability of the proposed LSTM-DQN attack framework to emerging CTC threats across diverse wireless environments.

Having established a robust detection framework, the next critical step is to exploit identified CTC signals to execute targeted jamming attacks that maximize disruption while maintaining stealth and efficiency. The challenge is to dynamically adapt jamming strategies to varying channel states and victim transmission patterns. This chapter presents a jamming attack model that integrates a LSTM network for predicting transmission probabilities with a DQN to optimize jamming decisions, enabling precise and energy-efficient interference in heterogeneous wireless environments.

B. Time-Slot Model

In the jamming attack framework, time is discretized into a sequence of fixed-length slots to maximize throughput reduction at the victim while balancing energy expenditure and stealth. We set each slot duration to 1 ms. During each slot t , the attacker obtains a CSI sample $\mathbf{x}_t \in \mathbb{R}^N$, where N denotes the number

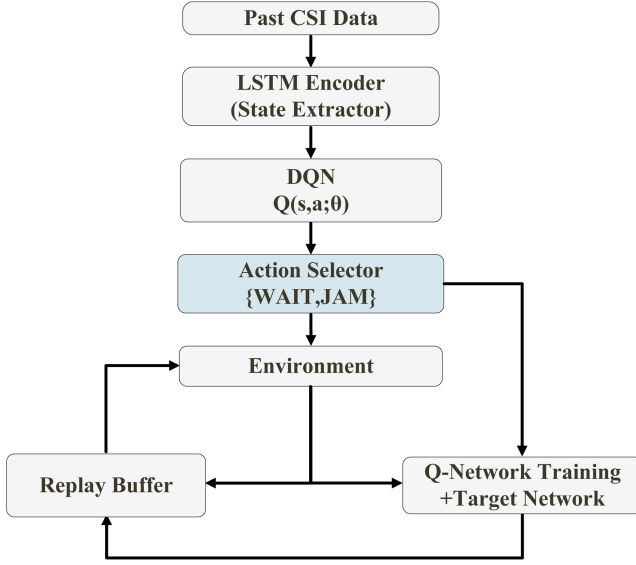


Fig. 6. The architecture of LSTM-DQN network.

of selected subcarriers. Assuming that the victim's traffic remains approximately stationary over short intervals, the attacker leverages the recent CSI sequence to predict the probability of transmission in the next slot, and thereby decides whether to launch a jamming attack. We partition the channel into contiguous 1ms slots indexed by i .

In slots where the victim transmits (e.g. slots $i + 1, i + 4, i + 8$), the attacker continuously monitors the CSI samples x_t . Once the attacker's LSTM-DQN pipeline predicts a high probability of imminent CTC transmission, it decides to jam or not in selected future slots (for example, slots $i + 2, i + 5, i + 9$ as shown). By staggering its jamming actions rather than continuously transmitting noise, the attacker maximizes disruption of the victim's sporadic transmissions while minimizing its own energy expenditure and reducing the risk of detection.

C. LSTM-DQN for Jamming Optimization

We design a neural network based framework that processes CSI data, predicts CTC patterns, and determines the optimal jamming time slots. The model integrates a LSTM network for CSI sequence modeling and a DQN for jamming decision optimization. The overall architecture of the proposed LSTM-DQN model is illustrated in Fig. 6. DQN combines deep learning with reinforcement learning to handle complex decision making tasks in high dimensional environments. While traditional Q-learning relies on a tabular Q-function that becomes impractical as the state-action space expands, DQN leverages a neural network to approximate the Q-function.

This enables efficient generalization in large and continuous state spaces. Within the MDP framework, DQN learns an optimal policy by maximizing the expected cumulative reward through experience replay and value iteration. In our jamming model, DQN is responsible for optimizing the jamming strategy based on real-time CSI observations. By learning from historical

interaction data stored in a replay buffer, the agent adapts to evolving traffic conditions and interference dynamics. Compared to rule based methods, DQN achieves stronger robustness and generalization by dynamically adjusting its behavior in response to environmental uncertainties, thus improving both energy efficiency and attack effectiveness.

Each time slot is set to 1 ms, consistent with the CSI sampling rate supported by commercial Wi-Fi hardware. At time slot t , the attacker acquires a CSI amplitude vector $x_t \in \mathbb{R}^N$, where N is the number of selected subcarriers. The input to the LSTM is the past k CSI samples, forming the sequence $\mathbf{X}_t = [x_{t-k+1}, \dots, x_t] \in \mathbb{R}^{k \times N}$. The LSTM outputs a hidden state h_t , from which a transmission probability prediction is generated as: $\hat{y}_{t+1} = \sigma(\mathbf{W}_o \mathbf{h}_t + \mathbf{b}_o)$, where $\hat{y}_{t+1}$ estimates the probability of transmission in slot $t + 1$, $\sigma(\cdot)$ is the sigmoid function, and $\mathbf{W}_o, \mathbf{b}_o$ are the output weights and bias. The agent constructs the MDP state vector $s_t \in \mathbb{R}^n$ by concatenating the most recent estimates of transmission probability n : $s_t = [\hat{y}_{t-n+1}, \dots, \hat{y}_t]^T$.

The binary action space is defined as $a_t \in \{0 = \text{WAIT}, 1 = \text{JAM}\}$, corresponding to whether to initiate jamming interference in the current time slot. The immediate reward is computed as:

$$r_t = \begin{cases} \alpha, & \text{if } a_t = 1 \text{ and } y_{t+1} = 1, \\ -\beta, & \text{if } a_t = 1 \text{ and } y_{t+1} = 0, \\ -\gamma, & \text{if } a_t = 0 \text{ and } y_{t+1} = 1, \\ 0, & \text{if } a_t = 0 \text{ and } y_{t+1} = 0, \end{cases} \quad (3)$$

where $y_{t+1} \in \{0, 1\}$ indicates the target's actual transmission in slot $t + 1$, and $\alpha, \beta, \gamma > 0$ respectively reward successful jamming, penalize unnecessary jamming, and penalize missed jamming opportunities. To learn the optimal policy, we define the Q-function as a neural approximation of the expected return:

$$Q(s, a; \theta) \approx \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s, a_t = a \right], \quad (4)$$

where θ denotes the parameters of the Q network and $\gamma \in (0, 1)$ is the discount factor. During training, the DQN minimizes the temporal-difference (TD) error. Transitions (s_t, a_t, r_t, s_{t+1}) are sampled uniformly from the replay buffer $\mathcal{D}$, and the TD target is calculated using a target network with parameters θ^- , so $y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-)$. The loss function is defined as the mean squared error between the TD target and the current Q-value estimate:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{D}} \left[(y_t - Q(s_t, a_t; \theta))^2 \right]. \quad (5)$$

The Q network consists of two fully connected hidden layers with ReLU activations, followed by an output layer that computes Q values for both actions. At each training iteration, the agent observes s_t , selects a_t , receives r_t , and transitions to s_{t+1} . The experience tuple is stored in $\mathcal{D}$, and the Q network is updated via a mini-batch gradient descent. To stabilize learning and mitigate overestimation of the Q-value, the target network $Q(\cdot; \theta^-)$ is synchronized with the main network $Q(\cdot; \theta)$ in every C step.

When DQN selects $a_t = 1$, the attacker transmits a predefined jamming signal during the time interval t , with a power level

at least five times greater than the ZigBee signal, sufficient to cause disruption while remaining within the Wi-Fi transmission limits. CSI monitoring continues after jamming, and known interference components are subtracted to determine whether CTC activity persists. If no target signal is detected, the system returns to the LSTM prediction and DQN decision cycle. Empirical results demonstrate that this LSTM-DQN architecture effectively reduces unnecessary jamming, enhances temporal precision, and significantly improves energy efficiency in real-time CTC attack scenarios.

VII. IMPLEMENTATION AND EVALUATION

A. Experimental Setup

To evaluate the performance of the proposed LSTM-Attention detection module and LSTM-DQN jamming framework, a real-world experimental testbed was established in the 2.4 GHz ISM band, simulating both normal CTC and adversarial interference scenarios involving Wi-Fi, ZigBee, and BLE devices. A desktop computer equipped with a commercial AC-1200 Wi-Fi network interface card (NIC), based on the RTL8812AU chipset [60], served as the CTC transmitter. This device dynamically switched between IEEE 802.11n and IEEE 802.11b modes to emulate ZigBee and BLE signals, respectively. On the receiver side, a Texas Instruments CC2650 [61] development board, supporting both ZigBee and BLE protocols, was configured using SmartRF Studio to capture cross-technology packets.

For jamming and CSI analysis, a second desktop running Ubuntu 6.8 was equipped with an Intel AX210S Wi-Fi card. This device acted as the intelligent attacker, performing passive CSI collection using the PicoScenes platform and executing detection and decision-making algorithms within a PyTorch 3.6 environment.

The training data used in this work was collected from a practical wireless testbed through controlled experiments. Specifically, the receiver continuously captured CSI measurements during different communication conditions, including Wi-Fi based CTC transmissions, normal ZigBee/BLE transmissions, and channel observations without CTC activity. To enhance the robustness of the dataset, data collection was performed under both LoS and NLoS environments, thereby covering representative channel variations encountered in practice. Each collected CSI sample was labeled based on the actual transmission state during the experiment, and a balanced dataset containing 3,000 samples was constructed, including 1,000 CTC samples, 1,000 standard ZigBee/BLE samples, and 1,000 non-CTC samples. The dataset was then split into 70% for training and 30% for testing.

Furthermore, for the jamming decision network, consecutive CSI observations were grouped into temporal sequences, so that the LSTM model could exploit inter-slot dependencies and dynamic channel evolution. This data collection and preprocessing procedure ensures that the trained model is based on realistic wireless measurements and can better reflect practical deployment conditions. To comprehensively assess system performance under diverse channel conditions, two representative deployment scenarios were designed:

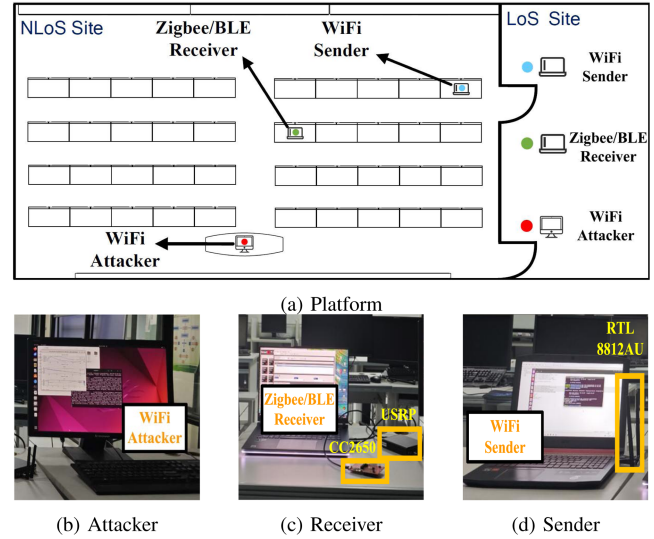


Fig. 7. Platform setting for indoor scenario.

- 1) *Non-Line-of-Sight (NLoS)*: A 10 m $\times$ 10 m indoor laboratory with furniture and obstructions, simulating multipath fading and shadowing effects.
- 2) *Line-of-Sight (LoS)*: A 3 m $\times$ 20 m corridor with minimal obstructions, providing strong direct path signals.
- 3) *Outdoor Open-Field*: An open outdoor environment (e.g., a campus road or parking lot) with minimal multipath reflections, utilized to evaluate the system's robustness against pure free-space path loss over extended distances (up to 50 m).

The experimental environment is depicted in Fig. 7. The upper portion illustrates the logical topology of the NLoS and LoS deployment sites, clearly indicating the relative positions of the transmitter, receiver, and attacker. The lower portion showcases the hardware setup, comprising the Wi-Fi transmitter with the RTL8812AU NIC, the CC2650 receiver, a USRP for passive spectrum monitoring, and the CSI enabled attacker node. To evaluate the effectiveness of our classification detection and attack model, we collected a dataset comprising 3,000 CSI samples: 1,000 containing CTC signals, 1,000 with standard ZigBee/BLE traffic, and 1,000 without any CTC activity. We partitioned the data into training (70%) and testing (30%) subsets.

B. Performance of LSTM-Attention Model

To evaluate the signal classification capability of the LSTM-Attention model under practical conditions, we conduct a series of experiments varying transmission power, distance, and packet length. These results reflect the model's robustness in classifying cross-technology signals across different propagation environments and protocol types.

Fig. 8 illustrate the accuracy trends for ZigBee and BLE in the outdoor scenario. The results indicate that the outdoor performance closely tracks the LoS scenario due to the clean channel conditions, and the LSTM-Attention model efficiently



(a) Outdoor scenario where ZigBee and BLE distributed in different locations

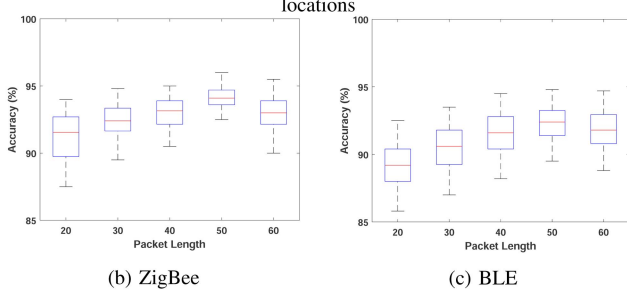


Fig. 8. The Accuracy under different package length in outdoor.

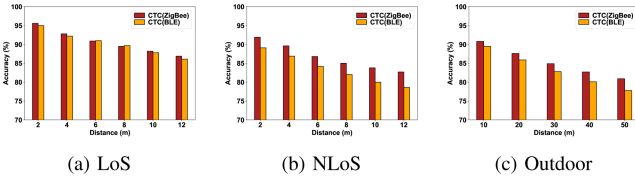


Fig. 10. The Accuracy under different distance in LoS, NLoS and outdoor scenes.

leverages longer packet lengths to consolidate temporal features, quickly converging to an accuracy of over 95%.

1) *Impact of Distance*: We then evaluate the effect of transmitter–receiver separation on model performance. As shown in Fig. 10, the accuracy consistently decreases with distance under both LoS and NLoS conditions. In the LoS case, ZigBee accuracy drops from 96% at 2 m to 87% at 12 m, while BLE decreases from 95% to 86%, showing a relatively smooth degradation. In contrast, the NLoS scenario exhibits a more pronounced decline, where ZigBee falls from 92% to 83% and BLE from 89% to 79%. Moreover, the performance gap between LoS and NLoS remains around 3%-5%, highlighting the impact of multipath and obstruction-induced signal distortion. Despite this, the model maintains over 80% accuracy even at longer distances under NLoS conditions, demonstrating its robustness in challenging propagation environments.

Furthermore, as shown in Fig. 10(c), the outdoor open-field scenario exhibits a smoother accuracy degradation trend due to the absence of severe multipath effects. Specifically, ZigBee accuracy decreases from 91% at 10 m to 81% at 50 m, while BLE drops from 89% to 78%. The performance gap remains stable within 1.3%-3.1%, and no abrupt fluctuations are observed, indicating that the degradation is primarily dominated by free-space path loss and demonstrating the model's robustness in outdoor environments.

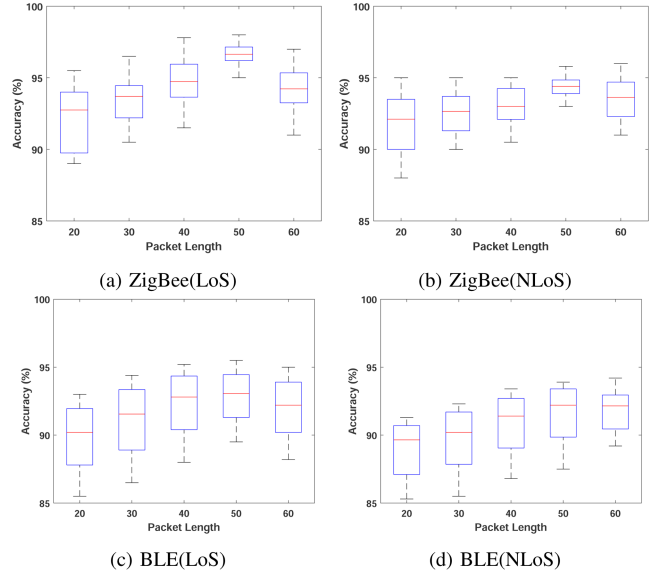


Fig. 9. The Accuracy under different package length in LoS, NLoS and outdoor scenes.

2) *Impact of Packet Length*: To explore the influence of temporal context, we vary the packet length and measure its impact on classification accuracy. As illustrated in Figs. 9 and Fig. 8, detection accuracy for both ZigBee and BLE signals exhibits a clear upward trend with increasing packet length. For instance, under Line-of-Sight (LoS) conditions, as the packet length increases, the baseline accuracy of ZigBee grows from 89% to a peak of 98%, while BLE similarly improves from 86% to 96%. This improvement highlights the effectiveness of temporal feature extraction from CSI sequences, as longer packets provide more stable and distinguishable patterns for classification. Furthermore, the performance under LoS conditions is generally higher than that under NLoS and outdoor scenarios due to reduced multipath interference and channel fading.

ZigBee signals demonstrate slightly better consistency and higher overall accuracy across different environments, benefiting from their fixed-spectrum DSSS modulation. Even in complex NLoS and outdoor environments, the maximum detection accuracy of ZigBee remains robust, reaching up to 96%. BLE, on the other hand, shows greater variability, as frequency hopping and channel distortion introduce fluctuating CSI patterns. Specifically, under NLoS conditions, BLE detection begins with a relatively lower accuracy of 86% at shorter packet lengths, but steadily improves to 94% as the packet length increases. A similar trend is observed in outdoor settings, where BLE accuracy climbs from 86% to 95%. These results reflect the model's ability to adaptively learn from increasingly informative temporal inputs, effectively mitigating environmental noise when sufficient CSI data is available.

3) *Model Comparison With State-of-The-Art Baselines*: To evaluate the advancement of our model, we compared it against classical ML models and a deep-learning based CSI detection baseline [33]. As shown in Table II, the LSTM-Attention model

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT DETECTION METHODS

Method	Accuracy	Precision	Recall	F1 Score
Random Forest	0.64	0.66	0.63	0.64
SVM	0.73	0.74	0.72	0.73
DNN-based CSI	0.89	0.89	0.88	0.88
LSTM-Attention	0.96	0.97	0.97	0.97

TABLE III
COMPARISON OF TIME CONSUMPTION FOR DIFFERENT MODELS

Model	T_p	T_c	T_u
LSTM-Attention	Pre-Trained	0.86 ms	—
LSTM-DQN	< 0.05 ms	0.92 ms	0.9 ms
Parallel Model	< 0.05 ms	0.92 ms	0.9 ms

achieves 0.96 accuracy and a 0.97 F1 Score, significantly outperforming all baselines.

While the DNN-based baseline improves accuracy to 0.89 over shallow models (SVM/RF), it still struggles to capture the fine grained temporal dependencies of CTC signals. In contrast, our dual-branch architecture extracts features from both amplitude and phase sequences. By leveraging the attention mechanism to focus on critical signal variations, our model achieves superior precision (0.97) and recall (0.97). This high reliability effectively minimizes false-triggering in downstream jamming, ensuring both the effectiveness and energy efficiency of the overall attack framework. The comparison results are presented in Table III.

C. Performance of LSTM-DQN Jamming Model

To evaluate the effectiveness of the proposed LSTM-DQN jamming strategy, we conducted experiments targeting ZigBee and BLE receivers under real-time CTC communication.

1) *Throughput Impact*: The goal of this experiment is to analyze the impact of intelligent jamming on the throughput performance of ZigBee and BLE communication systems. The throughput under normal communication and jamming conditions is measured to highlight the efficiency of the proposed LSTM-DQN jamming framework. As shown in Fig. 11, we measured the throughput performance of both receivers in two distinct phases: a baseline phase without interference (slots 0-7500) and a subsequent jamming phase (slots 7500 onward).

During the initial phase, both ZigBee and BLE maintain stable throughput close to 100%, demonstrating the reliability of CTC under normal conditions. Once jamming begins, throughput drops significantly for both technologies, confirming the disruptive effect of interference. Notably, the proposed LSTM-DQN based attack exhibits a sharper and more persistent degradation than the random jamming baseline. Under LSTM-DQN jamming, ZigBee throughput decreases by over 60%, while BLE drops by approximately 45%. In contrast, random jamming causes a milder reduction—about 40% for ZigBee and 25% for BLE. This performance gap highlights the temporal targeting precision of the proposed approach, which utilizes CSI-based transmission prediction to selectively jam critical time slots, avoiding unnecessary interference and maximizing

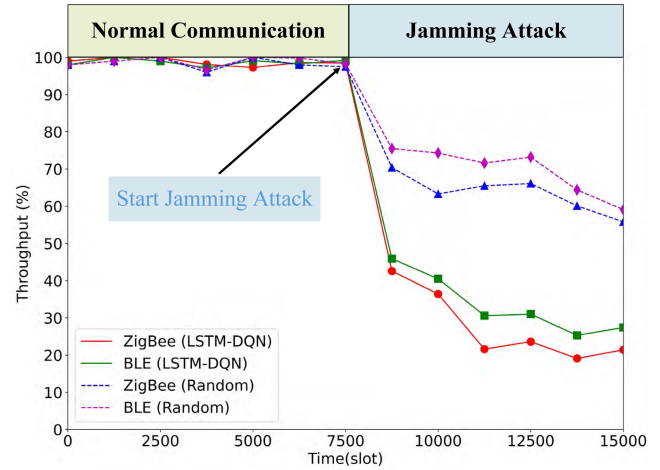


Fig. 11. Throughput Variations of ZigBee/BLE Under LSTM-Attention and Random Attack Scenarios.

attack efficiency. These findings demonstrate that the LSTM-DQN strategy outperforms conventional random jamming in both disruption strength and energy efficiency, validating its practical value for intelligent attacks on cross-technology links.

2) *Hit Rate Performance*: This experiment investigates the performance of the jamming model in terms of the Hit Rate (HR). The hit rate measures how well the jamming signal aligns with the victim's transmission, providing insight into the accuracy of the jamming strategy under different environmental conditions. To evaluate the effectiveness of the jamming model, we define the Slot-Level HR as an accuracy metric: $HR = \frac{N_{hit}}{N_{jam}}$, where N_{hit} denotes the number of time slots in which the jamming signal successfully overlaps with the victim's transmission activity, and N_{jam} is the total number of jamming attempts. Higher HR indicates that the jamming signal is better aligned with the actual transmission activity, resulting in more effective and energy-efficient interference.

Fig. 12(c) illustrates the HR values for ZigBee and BLE transmissions under LoS and NLoS conditions over varying distances from 1 to 11 meters. As expected, HR consistently decreases with increased separation between the attacker and the victim due to increased signal attenuation and reduced CSI prediction fidelity. In particular, LoS scenarios exhibit superior HR performance compared to their NLoS counterparts at all distances. For example, at a close range of 1 m, ZigBee achieves an HR of 92% under LoS conditions, while it drops to 86% in NLoS environments. As the distance extends to 11 meters, the HR for ZigBee gradually declines to 84% in LoS, but suffers a more severe drop to 74% in NLoS.

A similar pattern is observed for BLE transmissions. Under LoS conditions, the BLE hit rate maintains a relatively high level, decreasing from 91% at 1 m to 82% at 11 meters. However, in NLoS scenarios, the BLE hit rate experiences a steeper decline, starting at 83% at 1 m and falling below 80% beyond a distance of 3 meters, ultimately dropping to 72% at 11 meters.

This performance gap can be attributed to the more stable and predictable channel characteristics in LoS environments,

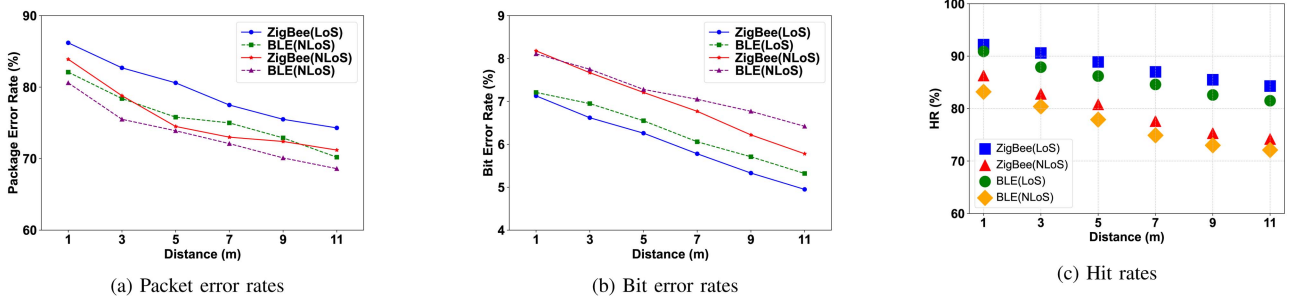


Fig. 12. The packet error rates, bit error rates and hit rates for ZigBee and BLE with different distances in LoS/NLoS.

which improve the accuracy of the LSTM-based CSI prediction model. In contrast, NLoS settings introduce multipath fading, reflection, and shadowing effects, all of which degrade CSI quality and consequently reduce the accuracy of predicted transmission probabilities. These results validate the importance of environmental awareness in CSI-driven jamming systems and underscore the necessity of adaptive decision strategies under dynamic propagation conditions.

3) *PER and BER Trends*: To assess the impact of intelligent jamming, we measured the Packet Error Rate (PER) and Bit Error Rate (BER) across distances from 1 to 11 meters (Fig. 12(a) and (b)). As shown in Fig. 12(a), PER decreases as distance increases due to signal attenuation. In LoS, ZigBee exhibits the highest PER, falling from 86% to 74%, while BLE drops from 82% to 70%. NLoS conditions offer improved resilience; for instance, BLE (NLoS) PER starts at 81% and declines to 69%. This suggests that multipath scattering in NLoS mitigates targeted jamming by introducing channel diversity.

Fig. 12(b) reveals that BER remains relatively low (5%-8%), confirming that the LSTM-DQN model primarily causes packet-level corruption rather than bit-level destruction. In LoS, both protocols see BER decrease from 7% to 5%. In NLoS, BER is slightly higher due to inherent multipath noise, starting at 8% and dropping to 6% at 11 meters. These results validate that while NLoS provides partial protection through channel variability, the proposed jamming remains highly effective at the packet level, particularly in close-range LoS environments.

4) *Comparison With Representative Approaches*: To comprehensively evaluate the advancement of our proposed system, we compare it against three representative jamming strategies: (1) SamBee [62], a state-of-the-art spoofing-based attack that fabricates legitimate preambles to hijack the channel; (2) LSTM-based Reactive Jamming [33], a recently proposed intelligent jammer that employs deep learning for signal detection followed by reactive interference; and (3) Random Jamming, which blindly transmits noise pulses without channel sensing.

Fig. 15(a) illustrates the jamming success rate as the distance between the attacker and the victim increases. While SamBee achieves high PER at close range, its effectiveness drops precipitously beyond 8 meters because the victim fails to decode the forged preambles under low SNR conditions. The LSTM-based reactive jammer also shows performance degradation at longer distances due to the inherent delay between detection and reaction, compounded by a higher missed-detection rate. Our

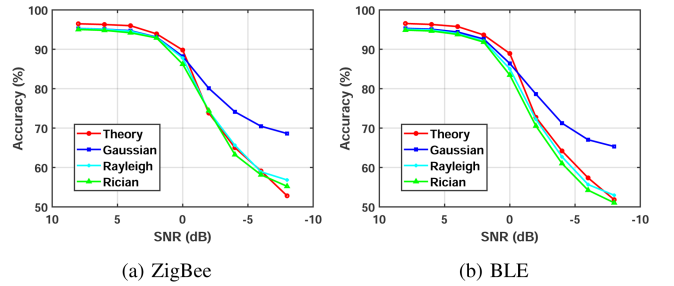


Fig. 13. The Accuracy for ZigBee and BLE under different SNR

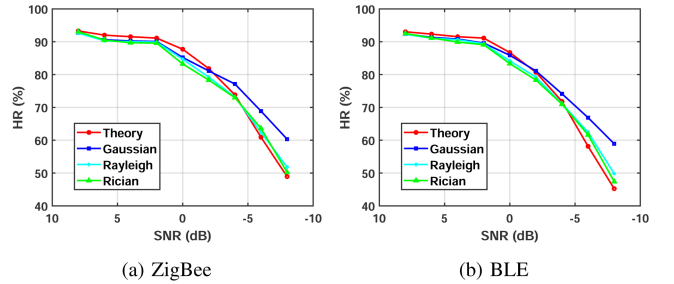


Fig. 14. The HR under different SNR

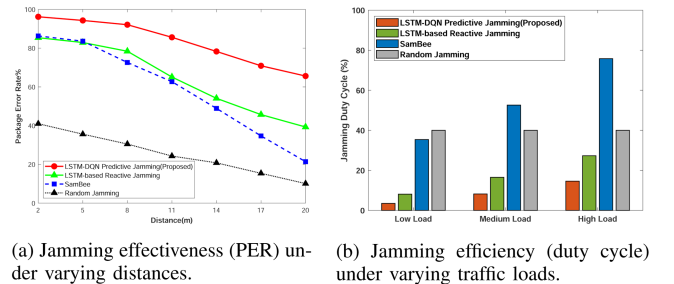


Fig. 15. Performance comparison with state-of-the-art jamming approaches.

proposed LSTM-DQN predictive jammer consistently outperforms the baselines. By forecasting the temporal transmission patterns, our system proactively aligns interference pulses with critical data slots, maintaining a robust PER of over 65% even at 20 meters.

In addition to effectiveness, Fig. 15(b) evaluates the energy efficiency of these jamming techniques by comparing their duty

cycle across varying traffic loads. SamBee requires the transmission of long forged packets to induce continuous channel backoff, resulting in an prohibitive duty cycle exceeding 70% under high traffic loads. The LSTM-based reactive jammer similarly consumes more energy to extend interference pulses and compensate for detection delays. In stark contrast, our LSTM-DQN framework significantly reduces energy consumption by strategically firing ultra-short interference pulses (e.g., 1-2 milliseconds) perfectly aligned with the predicted transmission slots. Consequently, it achieves a superior jamming success rate while maintaining a duty cycle below 15% even at high load, demonstrating a transformative leap in efficiency compared to existing intelligent jamming methods.

D. Impact of Channel Models and SNR Conditions

To assess the robustness of the proposed LSTM-Attention detection module and LSTM-DQN jamming framework under diverse wireless environments, experiments were conducted to evaluate performance across various channel conditions and noise levels. Specifically, classification accuracy and jamming hit rate were analyzed in additive white Gaussian noise (AWGN), Rayleigh fading, and Rician fading channels at different levels of SNR. CTC signals were generated using MATLAB, with synthetic channel models applied to the baseband waveform prior to over-the-air transmission via a USRP B210 in a controlled laboratory setting. The receiver captured CSI samples, which were processed by the LSTM-Attention and LSTM-DQN models.

Detection performance exhibited high robustness under AWGN, achieving a classification accuracy exceeding 95% for SNR levels above 4 dB. As illustrated in Fig. 13(a) and (b), accuracy gradually declined with decreasing SNR. In the fading channels, the performance degradation was more pronounced. For instance, at -6 dB, detection accuracy fell below 60% for Rayleigh and Rician channels, while remaining above 70% under AWGN conditions. Rician fading consistently outperformed Rayleigh at low SNR, likely due to its line-of-sight component providing more stable CSI features.

The jamming performance, measured by the hit rate, followed a similar trend. As shown in Fig. 14(a) and (b), the framework achieved a hit rate of 90% at high SNR in all channel models. Even at -6 dB, the hit rate remained above 70% under AWGN and approximately 58% under Rayleigh and Rician fading, demonstrating resilience in moderately degraded environments. These results validate the generalizability of the proposed framework. Although both detection and jamming performance degrade under severe channel conditions, the system maintains effective operation across diverse protocols and fading models, confirming its practical applicability in heterogeneous IoT scenarios.

E. Latency

This section evaluates the latency performance of the proposed LSTM-Attention signal detection model and the coordinated LSTM-DQN slot selection framework in CTC environments. Real-time responsiveness constitutes a critical

performance indicator, particularly in heterogeneous networks coexisting with Wi-Fi, ZigBee, and Bluetooth devices, where rapid signal detection and adaptive time slot allocation directly impact communication stability and security. To quantify the feasibility of the system for time sensitive applications, we measure three fundamental latency components: the data preprocessing time T_p required for CSI sample preparation, the model inference time T_c for generating prediction outputs and the online update time T_u necessary for the adaptation of dynamic parameters.

With each system timeslot configured to 1 ms duration, all computational processes including data processing, neural inference, and decision-making-must ideally complete within this constraints to ensure seamless real-time operation. However, achieving this requirement with computationally intensive deep learning architectures presents significant challenges. Following evaluation methodologies established in previous work like DeepJam [30], we define the real-time feasibility criterion as $\max(T_p, T_c, T_u) < 1\text{ms}$ and conduct experimental measurements in these latency dimensions.

1) *Latency of the LSTM-Attention Model for Detection:* For the pre-trained LSTM-Attention detection model, experimental focus centers on inference latency T_c given its direct impact on detection responsiveness. Hardware-accelerated timing measurements demonstrate stable end-to-end inference performance at 0.86ms, confirming compliance with single time-slot constraints. This performance benefits from offloading computationally intensive preprocessing operations T_p to offline pre-deployment stages, thereby minimizing real-time processing overhead.

2) *Latency of the LSTM-DQN Model for Decision-Making:* The LSTM-DQN decision-making model exhibits comparable real-time capabilities with measured maximum inference latency T_c reaching 0.92ms and asynchronous parameter update latency T_u peaking at 0.9ms under experimental conditions. State transition operations-converting detection outputs to DQN observational inputs-contribute negligible preprocessing delays ($T_p < 0.05\text{ms}$) due to their computational simplicity. The attacker's ability to perform the three tasks-data processing, inference, and model updating-in parallel is primarily attributed to two key factors. First, the experiment leverages a multi-core processor that supports parallel multitasking, enabling the efficient allocation of distinct tasks to separate cores for simultaneous execution. Second, the computational independence of these tasks, with minimal sequential dependencies between data processing, inference, and model updating, facilitates pipelined parallelism and asynchronous operations. Consequently, the decision module satisfies the requirement $\max(T_p, T_c, T_u) = T_c = 0.92\text{ms} < 1\text{ms}$, allowing a timely response to detected CTC activities.

3) *Latency of the Parallel Model Architecture:* Although sequential execution of the detection and decision modules would theoretically incur a cumulative latency of 1.78 ms, the practical deployment leverages pipeline parallelism to maintain sub-millisecond system responsiveness. In time slot t , the detection module processes CSI data and generates activity recognition results; in the subsequent slot $t+1$, the decision

module computes the jamming strategy based on the prior detection output. This staggered execution pattern temporally decouples detection from decision making, ensuring that each critical operation completes within a 1 ms slot. Experimental validation confirms that the system maintains near real-time performance, with a worst-case latency of 0.92 ms, even when considering inter-stage dependencies.

Although the proposed model achieves strong performance in the reported experiments, we acknowledge that the current dataset and evaluation scope have several limitations. Specifically, the classification model is trained and tested on a dataset of 3,000 CSI samples collected from our own testbed, with 70% used for training and 30% for testing. To reduce overfitting, we apply dropout regularization and select hyperparameters through grid search with five-fold cross-validation. Moreover, the model is evaluated under multiple channel conditions, including LoS and NLoS deployments, different transmit powers, different communication distances, different packet lengths, and synthetic AWGN, Rayleigh, and Rician channel settings across a range of SNR levels.

The comparison of time consumption for different models is presented in Table III. These results provide evidence that the learned CSI features are not tied to only one fixed channel realization. However, we agree that this evaluation still does not fully establish broad generalization across heterogeneous hardware platforms, different CSI acquisition chipsets, or independently collected datasets. In particular, the current experiments are based on a limited set of devices, including the RTL8812AU Wi-Fi NIC, TI CC2650 platform, USRP-based monitor, and AX210S-based attacker, and therefore should be interpreted as a real-world proof of concept rather than a definitive cross-platform benchmark.

VIII. DISCUSSION

Although this work focuses on CSI-driven detection and intelligent jamming against CTC transmissions, it also suggests several possible defense directions. First, since the attacker relies on temporal CSI regularity to infer the target's transmission behavior, one practical countermeasure is to introduce controlled randomness into transmission timing, packet intervals, or waveform scheduling, thereby reducing the temporal predictability exploited by the LSTM-based inference module. Second, dynamic channel switching and frequency-agile communication can reduce the success probability of slot-level jamming by weakening the attacker's ability to maintain stable monitoring and synchronized interference.

Third, adaptive transmit-power control and redundancy-aware retransmission mechanisms may further improve resilience by allowing legitimate receivers to recover from partial interference. In addition, CSI-based jammer detection can be deployed on the victim side to identify abnormal channel perturbations, followed by defensive actions such as channel migration, transmission rescheduling, or temporary fallback to more robust communication modes. More broadly, machine-learning based jamming classification and reactive-jamming countermeasures reported in prior studies can be incorporated into future CTC systems to build an attack-aware and self-adaptive defense

framework. These directions are beyond the scope of the current paper, but they represent important next steps toward securing heterogeneous cross-technology wireless communications.

IX. CONCLUSION

This paper systematically investigates the identification and intelligent jamming of cross-technology signals in heterogeneous wireless environments, proposing an attack framework that integrates deep learning and reinforcement learning. A dual-branch LSTM network with an attention mechanism is developed to achieve high-accuracy classification of Wi-Fi, ZigBee/BLE, and CTC signals. Additionally, an LSTM-DQN model is employed to optimize slot-level jamming decisions. Experimental results demonstrate the system's robustness and real-time performance across diverse channel conditions, achieving a jamming success rate exceeding 90% in LoS scenarios and significantly degrading ZigBee/BLE communication performance. The framework's adaptability is further validated under AWGN, Rayleigh, and Rician channel models, underscoring its practical value in complex wireless environments.

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